

MODEL VALIDATION, OVERFITTING ANALYSIS, AND HYPERPARAMETER TUNING REPORT

Task Assignment: Task-3 (Advanced Model Evaluation & Optimization)

Project System: House Price Prediction System

Dataset utilized: California Housing Dataset

Target Variable: Median House Value

1. Introduction & Executive Summary

This report details the technical implementation and analytical findings for Task-3 of the Machine Learning Engineering Internship. Building upon the baseline linear models established in Task-2, this phase introduces rigorous evaluation structures to diagnose model variance, handle non-linear feature relationships, and systematically optimize structural hyperparameters.

By migrating from a simple validation split to an industry-standard cross-validation workflow, we ensure the final deployed model generalizes robustly to unseen real estate data.

2. Overfitting Analysis & Variance Diagnosis

To establish a performance baseline for tree-based architectures, an unconstrained **Decision Tree Regressor** was trained on the standardized training data. An unconstrained tree splits nodes continuously until all training samples reside in pure leaf nodes.

Upon evaluating the model on both splits, the following Root Mean Squared Error (RMSE) metrics were observed:

- **Training Set RMSE:** 0.0000 (Perfect historical fit)
- **Testing Set RMSE:** 0.7321

Technical Interpretation:

The massive numerical gap between the training performance and testing performance is textbook evidence of **high variance (overfitting)**. Because the algorithm faced no structural depth constraints, it memorized individual data points and spatial noise unique to the training sample instead of learning universal patterns. While it exhibits zero error on historical data, its poor generalization boundaries make it structurally unviable for production readiness without optimization.

3. Hyperparameter Tuning Methodology

To counteract the observed overfitting, automated optimization was performed utilizing scikit-learn's GridSearchCV module. This process enforces structural regularizations on the tree layout by testing a parameterized grid matrix:

Selected Parameter Space:

- `max_depth`: [3, 5, 7, 10] — Restricts the maximum vertical depth of the tree, forcing the model to create broader, more generalized rules.
- `min_samples_split`: [2, 5, 10] — Restricts further node partitioning unless a minimum threshold of samples is present, effectively smoothing out outliers.

The tuning architecture evaluated all 12 unique combinations across a 5-fold cross-validation scheme, assessing a total of 60 individual training iterations to converge on the mathematically optimal structural layout.

4. Methodological Justification: Why Cross-Validation is Trusted

Relying strictly on a single train-test split introduces statistical risk due to the "luck of the draw." If a single random partition accidentally groups anomalous luxury properties or distinct geographic clusters entirely into the test subset, performance metrics skew negatively and present an inaccurate picture of model health.

The 5-Fold Cross-Validation Framework:

Cross-Validation addresses this limitation by partitioning the complete dataset into 5 equal, distinct subsets (folds). The model undergoes 5 separate training cycles; in each cycle, 4 folds act as the training set, while the remaining 1 fold serves as the validation gate.

Because every single data point is guaranteed to serve as both a training element and an unseen validation element, the resulting averaged metric is structurally stable. This removes sample bias and yields a realistic performance estimation that data professionals trust before deployment.

5. Model Comparison & Performance Synthesis

After completing the optimization loop, the tuned model metrics were compiled alongside the baseline statistics established during Task-2 for comparative evaluation:

Final Performance Summary Table:

Model Architecture	Evaluation Framework
Linear Regression	Train-Test split

Ridge Regression	Train-Test split
Tuned Decision tree	5 – fold cross validation

6. Final Model Selection Justification & Engineering Trade-offs

Definite Model Selection:

The Tuned Decision Tree is selected as the optimal production model for the House Price Prediction system.

Justification Matrix & Trade-offs:

Mathematical Superiority: Linear and Ridge regression models structurally underperformed, maintaining a higher error rate (RMSE: ~0.7455) and lower predictive power (R^2 : 57.58%). Linear models assume straight-line interactions, rendering them incapable of capturing complex geographical boundaries and non-linear multi-variable interactions inherent in California housing data (such as proximity to clusters like San Francisco or Los Angeles).

Mitigation of Overfitting: By introducing structural constraints via GridSearchCV, the Tuned Decision Tree successfully avoided the overfitting loop of the base tree, driving the model to a highly competitive, generalized RMSE of 0.6725.

Complexity vs. Interpretability Trade-off: Selecting a non-linear tree-based architecture over a basic linear model requires a conscious trade-off regarding simplicity. Linear models output simple formulas with direct multipliers. A decision tree creates deep branches that are slightly more challenging to audit manually. However, by capping the `max_depth` to a stable maximum of 10, we preserve clear, chartable decision paths while capturing non-linear pricing logic, providing an optimal balance between interpretability and predictive accuracy.